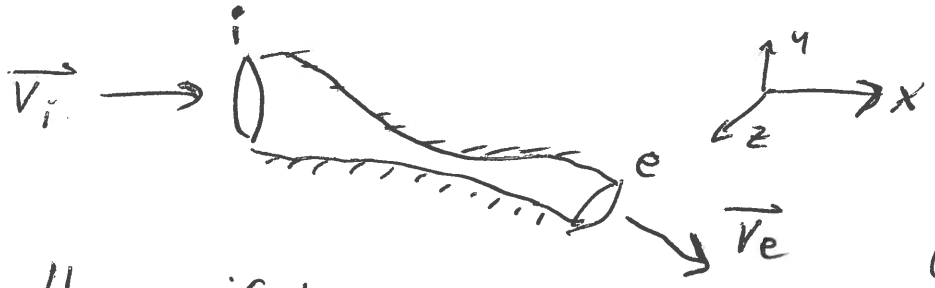
Brief review of flow modeling/engine modeling issues from Propulsion 1 & fluids

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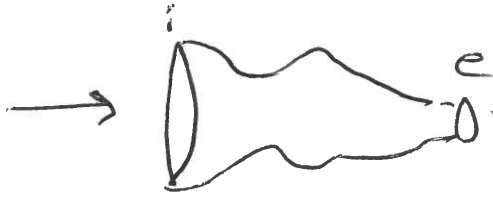
Quasi-1-D flow model - flow in a streamtube modeled as having uniform properties ($|\vec{V}|$, P, T, $\rho$, etc.) across a given cross-section ($\perp$ to local flow direction). Can provide very useful info for both jet & rocket engine flows. (Know the limitations, though!)

For an engine (or a section of engine) modeled using quasi-1-D flow, with changes in flow direction between i & e, the momentum equation is a vector equation, so $F_x$, $F_y$ must be assessed & are based on angularity / changes in flow direction. Note $F_x$, $F_y$ (say from eq. 11 and eq. 12) are the net force components on all (internal) side surfaces between i & e.




Assess $F_x$, $F_y$, $F_z$ using C.V. Analysis (eq. 11 & eq. 12) & eq. 13

However, if there is a 'straight-line' predominant direction to the flow between i & e (such that the flow velocity can be fully characterized by a Scalar 'u' throughout, say in the 'local' x direction)




(single)

then $F_x = -\{\dot{m}_e V_e - \dot{m} V_i + P_e A_e - P_i A_i\}$

& $F_y (\& F_z) = 0$, by definition, if all internal side surfaces of streamtube are 'wetted surfaces' → you show it!

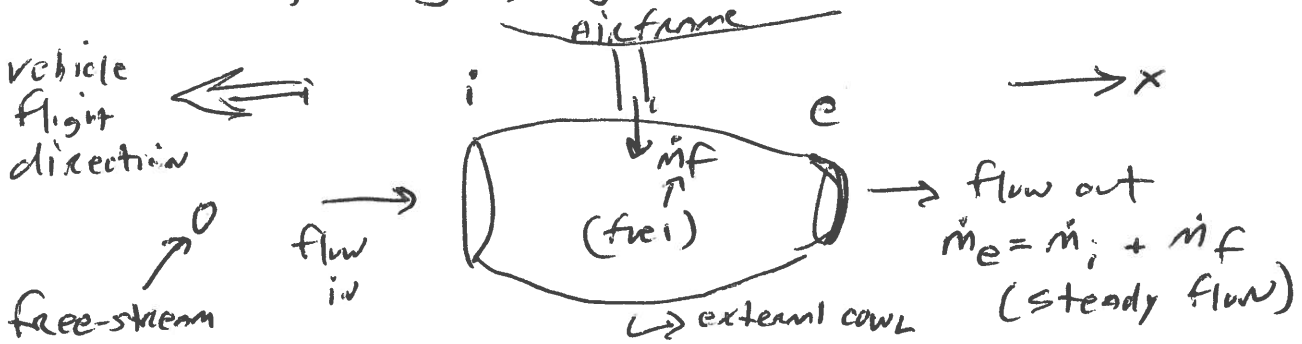
Note this is written as

$$|F_x| = \Delta ST \quad (ST = \text{'Stream thrust'})$$

$\Delta ST = ST_e - ST_i$ where $ST = \dot{m} V + PA$

---

Hence, for a (jet) engine (with such modeling)




Obviously, a positive (+) 'thrust' corresponds to a net negative force in the x direction (for orientation shown).

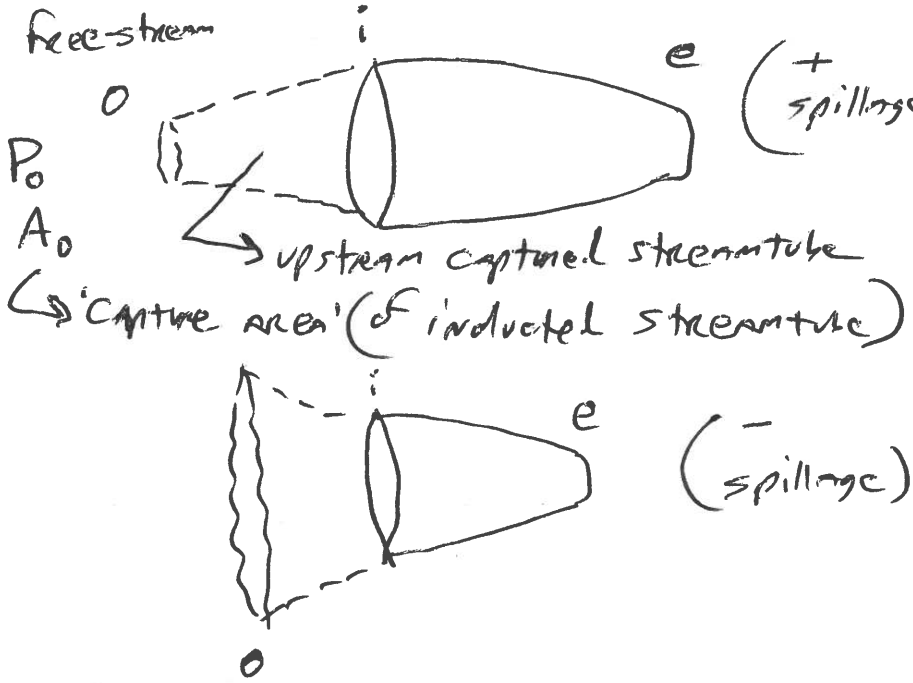
"Engine Installed Thrust" is the actual force experienced by the engine & transmitted up the strut to airframe:

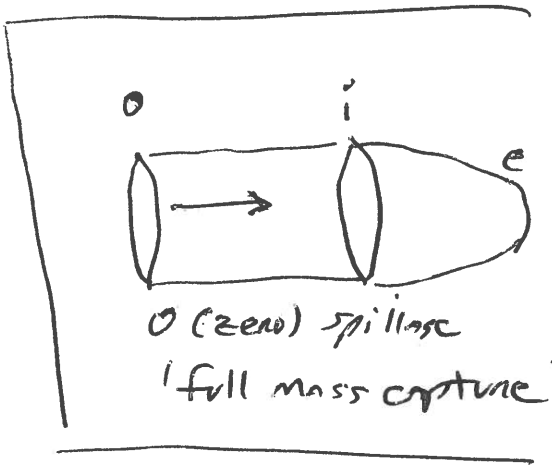
$$= |F_x| - D_{ext}$$

$D_{ext}$ = drag on external cowl of engine due to pressure (Note drag is + in + x direction)

[viscous external cowl force assessment charged to airframe force accounting (traditionally)]

Now, consider the streamtube inducted & processed by the engine; it can be curved upstream of the engine (between 0 & i)




[ The degree of upstream curvature (+, −, how much) depends on the engine operating condition ($\dot{m}_f$, ALT, $M_0$) for a given engine. Note if $M_0 > 1$ (supersonic flight), ANY curvature upstream of i implies subsonic flow entering the engine (since supersonic flow has, by definition, straight streamlines). (So that would imply a shock system upstream of i to decelerate the flow from supersonic to subsonic, upstream of i) ]

---

The 'textbook definition' of engine thrust is given as

('Uninstalled') Thrust $= \dot{m}_e V_e - \dot{m}_0 V_0 + (P_e - P_0) A_e$

(note $\dot{m}_0 = \dot{m}_i$, by definition)

But this definition of thrust is properly called the 'Uninstalled thrust' and is based on assumption of

constant $P_0$ (ambient pressure) on external cowl of engine from i to e and 'ignores' axial force on captured streamtube from 0 to i ... Note there will be an axial force on the captured streamtube from 0 to i if streamtube is curved (+ or - spillage) → due to pressure on sides of streamtube ...

So ...

$$\text{Installed Thrust} = \text{Uninstalled Thrust} - D_{ext} - D_{add}$$

where $D_{ext}$ = external cowl drag due to $P \neq P_0$ on external cowl (i.e., when engine is in flight!)

$D_{add}$ = 'additive' drag due to spillage upstream of engine

Warning: an ignorant engineer, who has not thought carefully about this (or has not studied), could (for example) simply find/use the standard 'textbook' definition of engine (uninstalled) thrust, give that number to vehicle analyst who is doing the force accounting ... Result would be incorrect actual force experienced by vehicle!

Don't be an ignorant engineer

Be very methodical & logical about 'force accounting'!

For rocket engines:

There is no air inlet for a rocket.

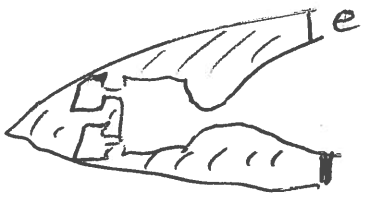
$F_x$ is the net 'axial' force on all internal wetted surfaces of the engine out to 'e' [(exit plane of nozzle) including fuel/oxidizer feed & tankage systems]

so $F_x = \dot{m} V_e + P_e A_e$, by definition...

But, again, the 'textbook definition' of rocket thrust is given as

$$\text{Thrust} = \dot{m} V_e + (P_e - P_o) A_e$$

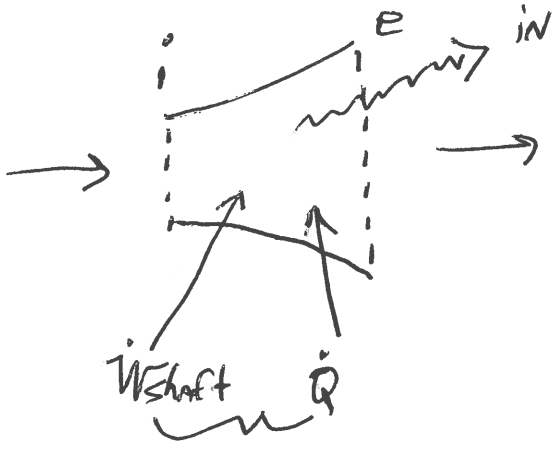
But this assumes a constant $P_o$ on all external surfaces of the rocket




---

Understand and correctly implement accurate 'force accounting'

# Total Conditions Review (quasi-1-D flow)




internal 'losses' (irreversibilities) like friction, shocks, etc.

[CPG gas model (calorically perfect gas) $\gamma, C_p, R$ constant]

$\dot{E}$ (J/sec or Watts)

$P_t, T_t$ → 'total' conditions at a station...  ($\rho_t = \frac{P_t}{RT_t}$, etc.)

'Reference' conditions; pressure & temp. obtained if flow slowed isentropically to zero velocity. (AND NO external work!)

Reminder: 'isentropic' means constant entropy; so an isentropic process requires Adiabatic (no heat transfer across boundary) & No 'losses' (no friction, shocks, etc.)

---

How do $P_t, T_t$ VARY in a streamtube? (i.e., What 'drives' changes in them from one station to another?)

Define 'work per mass' & 'heat per mass' in a quasi-1-D flow (interactions)

$$w_{i\to e} = \frac{\dot{W}_{i\to e}}{\dot{m}} \quad \& \quad q_{i\to e} = \frac{\dot{Q}_{i\to e}}{\dot{m}}$$

$$w_{i\to e} = \int_i^e \delta w \qquad q_{i\to e} = \int_i^e \delta q$$

The energy equation (applied between i & e) is simply

$$\dot{m}\left(c_p T_e + \frac{v_e^2}{2}\right) - \dot{m}\left(c_p T_i + \frac{v_i^2}{2}\right) = \dot{Q}_{i\to e} + \dot{W}_{i\to e} \left(= \dot{E}\right)$$

OR

$$\left(c_p T_e + \frac{v_e^2}{2}\right) - \left(c_p T_i + \frac{v_i^2}{2}\right) = q_{i\to e} + w_{i\to e}$$

OR in terms of $T_t$ ...

$$c_p(T_{te} - T_{ti}) = q_{i\to e} + w_{i\to e}$$

or

$$\frac{T_{te}}{T_{ti}} = 1 + \frac{q_{i\to e}}{c_p T_{ti}} + \frac{w_{i\to e}}{c_p T_{ti}}$$

Simply the energy equation ...

Recall

$$\frac{T_t}{T} \simeq 1 + \frac{\gamma-1}{2} M^2$$

at any station & enthalpy is $c_p T$ so total enthalpy (per mass) is $c_p T_t$ !

$T_t$ can be shown that (not done here)

$$\frac{P_{te}}{P_{ti}} = e^{\left[\int_i^e \frac{dsw}{RT_t} - \frac{s_{IRR/NON-ideal\ heat\ transfer,\ i\to e}}{R}\right]}$$

$\frac{J}{1\text{-}Hg}$

$s_{IRR}$ → entropy (per mass) due to internal irreversibilities (like friction, shocks)

$s_{NON\ ideal\ h.t.}$ → occurs if heat interaction ($\dot{Q}_{i\to e}$) where Mach of flow $M \neq 0$ (i.e., $M > 0$); higher M, the more $s_{NON-ideal\ ht}$ !

Aside: does this make sense for (say) an isentropic compression? (check)

By definition, an isentropic compressor has no $S_{IRR}$, no $S_{NON-IDEAL}$ (no $q$!)

So from energy equation: $C_p dT_t = \delta w$ (no $\delta q$) across a differential step, i.e.

$$\left(\frac{P_{te}}{P_{t_i}}\right)_{\substack{\text{isentropic} \\ \text{compression}}} = e^{\int_i^e \frac{C_p dT_t}{RT_t}} \quad \text{but } \frac{C_p}{R} = \frac{\gamma}{\gamma - 1} \text{ (identity)}$$

or

$$\left(\frac{P_{te}}{P_{t_i}}\right)_{\substack{\text{isentropic} \\ \text{compression}}} = \left(\frac{T_{te}}{T_{t_i}}\right)^{\frac{\gamma}{\gamma-1}}$$

---

Hence ... total temp, $T_t$, changes in a flow are due <u>only</u> to external energy interactions ($\dot{Q}$ or $\dot{W}$), and changes in $T_t$ do not distinguish between heat and work interactions! (so $T_t$ is the measure of the 'total energy content' of a flow).

$P_t, T_t$ are 'thermodynamic handles'. They don't change just because flow is expanding or contracting (cross-sectional area) unless $\dot{Q}, \dot{W}$, internal irreversibility! (static $P, T$ do!) ... so very useful!

<u>total pressure, $P_t$</u>, changes in a flow are due to work interactions (+ work as in a compressor raises $P_t$, − work as in a turbine lowers $P_t$)

<u>AND</u> (always!)

$P_t$ <u>drops</u> for any irreversibility <u>And</u> in (+) non-ideal heat transfer (where $M$ of flow $> 0$)

# STREAMTUBE FLUID DYNAMICS (Review)

## Mass flow rate and area relationships in quasi-1-D flow

$$\dot{m} = \rho U A = \frac{P}{RT} M\sqrt{\gamma RT}\, A = P_t \sqrt{\frac{\gamma}{R}} \frac{1}{\sqrt{T_t}} MA\left(1+\frac{\gamma-1}{2}M^2\right)^{\frac{-(\gamma+1)}{2(\gamma-1)}} \quad (*)$$

$$\left\{ P_t = P\left(1+\frac{\gamma-1}{2}M^2\right)^{\frac{\gamma}{\gamma-1}} ; \quad T_t = T\left(1+\frac{\gamma-1}{2}M^2\right) \right.$$

So: (*) gives Mach # ($M$) at a station in a streamtube, provided $\dot{m}$, $P_t$, $T_t$, $A$ known (must solve for $M$)

Note: 2 solutions exist (sub & supersonic) → 2 roots ∴ which depends on fluid situation

---

Apply (*) at a throat (* → throat) — could be a reference throat, not a physical throat, where $M=1$ and $A = A^*$:

$$\dot{m}^* = P_t^* A^* \sqrt{\frac{\gamma}{RT_t^*}} \left(\frac{\gamma+1}{2}\right)^{\frac{-(\gamma+1)}{2(\gamma-1)}}$$

exponent → $\frac{\gamma+1}{2(\gamma-1)}$

Now define $C^*$ ('characteristic velocity'): $C^* = \sqrt{\frac{RT_t^*}{\gamma}\left(\frac{\gamma+1}{2}\right)^{\frac{\gamma+1}{\gamma-1}}}$

then $\dot{m}^* = \frac{P_t^* A^*}{C^*}$ ; Note $C^* = C^*(T_t^*, \gamma, R)$

If flow is steady in quasi-1-D streamtube $\dot{m} = \dot{m}^*$

If flow is isentropic (adiabatic (→ no heat interaction) & reversible (→ no losses ($\dot{S}_{ge}$)), steady & no external work interaction, then $P_t, T_t$ are const (between a location of interest & throat * where $A = A^*$, $M = 1$ (Again, whether actual or reference throat

Then, since $\frac{\dot{m}}{\dot{m}^*} = 1.0$:

$$\boxed{\frac{A}{A^*} = \frac{1}{M}\left[\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}} = G(M)}$$

Although this $\left(\frac{A}{A^*}\right)$ is tabulated for $\gamma = 1.4$ in 'isentropic flow tables' versus Mach # $M$ (for both sub & supersonic locuses), it is best to use the equation (find the root) since often in engines $\gamma \neq 1.4$.

If flow is isentropic (no $\dot{Q}$, no losses) & no work interaction ('$\dot{W}_{shaft}$') from surroundings, then

$P_{t_2} = P_t^* = P_{t_1}$ & $T_{t_2} = T_t^* = T_t$, & $A_2^* = A^* = A_1^*$

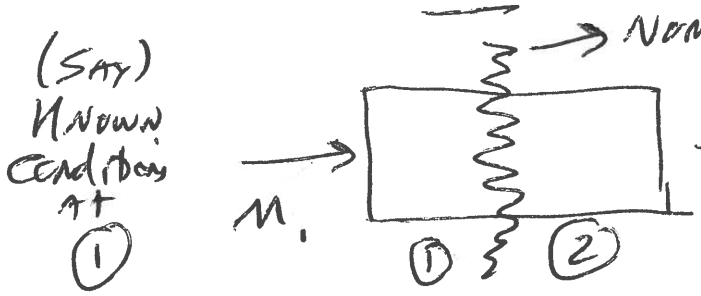
so $\boxed{A_2/A_1 = \frac{G(M_2)}{G(M_1)}}$; hence given $\frac{A_2}{A_1}$ & $M_1$, you can find $M_2$ (or any variation)

Also, if no $\dot{Q}$, $\dot{W}$ interaction between 1 & 2 then $C^* =$ constant (since $T_t$ stays constant)

so $\boxed{P_{t_1} A_1^* = P_{t_2} A_2^*}$ Good for non-isentropic flow (but no external energy interactions $\rightarrow \dot{Q}, \dot{W}$)

## Normal Shocks:

Apply 1-D control volume ("differential" C.V.) analysis thru a NORMAL shock




$\Rightarrow$ C.V. yields

$$\frac{dp}{\rho} + \frac{dv}{v} = 0$$

$$\frac{dP}{\rho} + v\,dv = 0$$

$$c_p T + v\,dv = 0$$

$$P = \rho R T$$

These differential equations can be solved (analytically) to yield

$$M_2^2 = \frac{M_1^2 + \frac{2}{\gamma-1}}{\frac{2\gamma}{\gamma-1}M_1^2 - 1} \qquad \frac{P_2}{P_1} = \left(\frac{2\gamma}{\gamma+1}\right)M^2 - \frac{(\gamma-1)}{(\gamma+1)}$$

NO energy interactions from surroundings (no $\dot{Q}, \dot{W}$)

$T_{t_1} = T_{t_2}$ (but since shock is irreversible $\rightarrow$ losses)

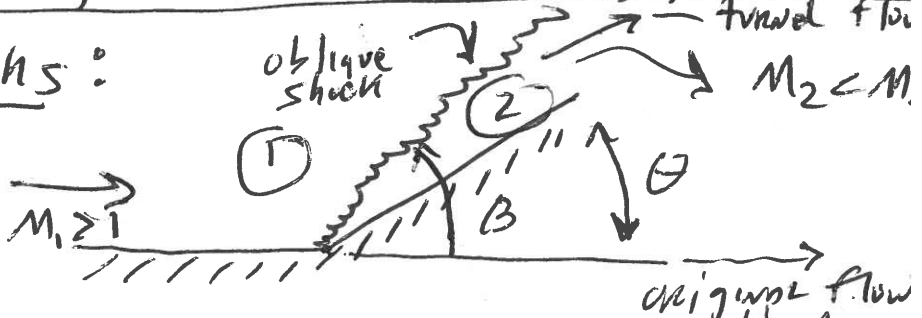
(NON-isentropic $P_{t_2} < P_{t_1}$)

Can find $T_2$, then $v_2 (= M_2\sqrt{\gamma R T_2})$, etc.

---

## Oblique Shocks:

(2-D)




$\theta$ = deflection (turning angle) of flow

Region 2 is uniform region of turned flow (compression)

Analysis shows:

$$\tan\theta = 2\cot\beta\left[\frac{M_1^2\sin^2\beta - 1}{M_1^2(\gamma + \cos 2\beta) + 2}\right]$$

$\theta$-$\beta$-M

Rel

(Anderson)

(oblique shocks, cont.)

So, for known $\theta$, $M_1$, $\gamma$, Find $\beta$ that satisfies the $\theta-\beta-M$ relation.

Solve this yourself → do not rely on oblique shock chart!!!

Reminder:
oblique shock chart plots $\theta-\beta-M$ relation for $\gamma = 1.4$
'strong shock solution'
'weak shock solution'

(CANNOT – even remotely – read chart, especially for high Mach #'s !!!)

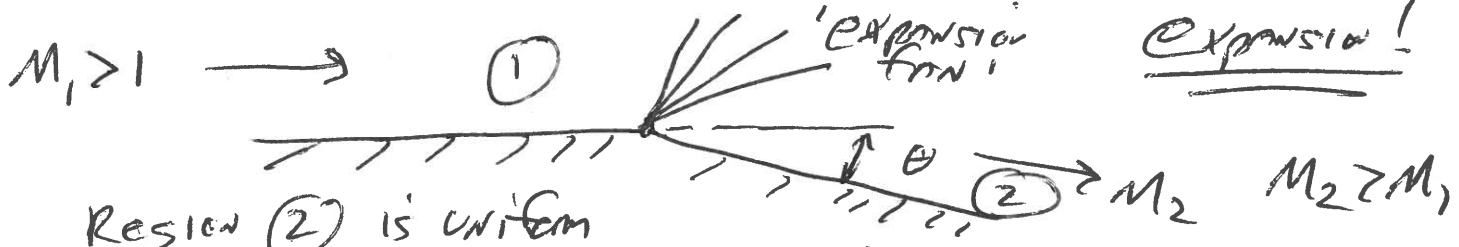
Then $M_2 = \dfrac{M_{N2}}{\sin(\beta-\theta)}$ where $M_{N2}^2 = \dfrac{M_{N1}^2 + \left(\frac{2}{\gamma-1}\right)}{\left[\frac{2\gamma}{\gamma-1}\right]M_{N1}^2 - 1}$

& $M_{N1} = M_1 \sin\beta$ and $\dfrac{P_2}{P_1} = 1 + \left[\dfrac{2\gamma}{\gamma-1}\right]M_{N1}^2 - 1$

& since $T_{t2} = T_{t1}$, find $T_2, \rho_2, P_{t2}$, etc...

---

Prandtl-Meyer Expansion fans (inviscid)




Region (2) is uniform region of turned flow (expanded)

isentropic (no coalescence of Mach waves → they spread out)

$\theta$ = turn angle

So $\boxed{P_{t2} = P_{t1} \quad T_{t2} = T_{t1}}$

## Prandtl-Meyer Expansion fan (cont.)



Analysis shows:

$$\theta = \nu(M_2) - \nu(M_1)$$

where $$\nu(M) = \sqrt{\frac{\gamma+1}{\gamma-1}}\arctan\sqrt{\frac{\gamma-1}{\gamma+1}(M_1^2-1)} - \arctan\sqrt{M_1^2-1}$$

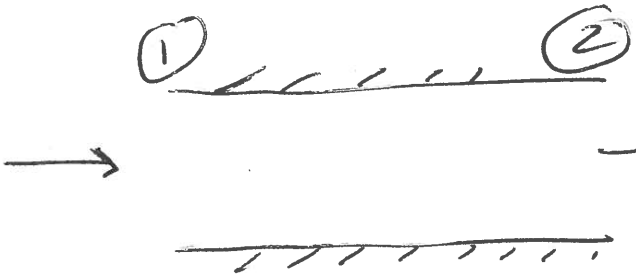
So: find $M_2$ (for given $M_1, \theta$) that satisfies this equation! (Again, do not use tables, especially for high speed!)

Then find $T_2, P_2, \rho_2, V_2$ etc. (since $P_t, T_t$ con

---

**Rayleigh flow** (1-D, frictionless, no area variation, no work interaction) but with heat addition (NOW - adiabatic)

1-D → no area change!




Energy equation results in

$$c_p(T_{t_2} - T_{t_1}) = q_{1\to 2}$$

$$q_{1\to 2} = \frac{\dot{Q}_{1\to 2}}{\dot{m}} \quad \left(\frac{J}{kg\ air}\right)$$

Analytical solution for this kind of flow shows

$$\frac{T_{t_2}}{T_{t_1}} = \frac{f(M_2^2)}{f(M_1^2)}\left(= 1 + \frac{q_{1\to 2}}{c_p T_{t_1}}\right)$$

where $$f(M^2) = \left\{\frac{1+\frac{\gamma-1}{2}M^2}{(1+\gamma M^2)^2}\right\}M^2$$

## Rayleigh Flow (cont.)

So if $q_{1\to 2}$ known (& $T_{t_1}$, $M_1$ known)
find $f(M_1^2)$, then find $f(M_2^2)$ & then solve for $M_2$ from:

$$M = \sqrt{\frac{2f}{1-2\gamma f \pm \left[1-2(\gamma+1)f\right]^{1/2}}}$$

+ subsonic flow
− supersonic flow

& $\frac{P_2}{P_1} = \frac{1+\gamma M_1^2}{1+\gamma M_2^2}$ & since $M_2$ known, now find all other conditions at (2) (or any variance!)

→ There are $P_t$ losses ('Rayleigh losses' due to q added at $M \neq 0$ & these $P_t$ losses get larger the bigger M is !! (quite large for supersonic flow)

---

'Fanno Flow': 1-D, adiabatic, with friction
[you review!]

---

**Engine and thermal choking (& friction choking)**

See your AE 1535 notes!!

Rayleigh flow shows the 'thermal choking' issue very well; heat addition (say exothermic model for fuel combustion) ALWAYS drives Mach toward 1.0 (on either subsonic or supersonic legs)

If you try to add more heat anyway (beyond thermal choking), conditions at (1) will necessarily change! (different problem)

AE5535

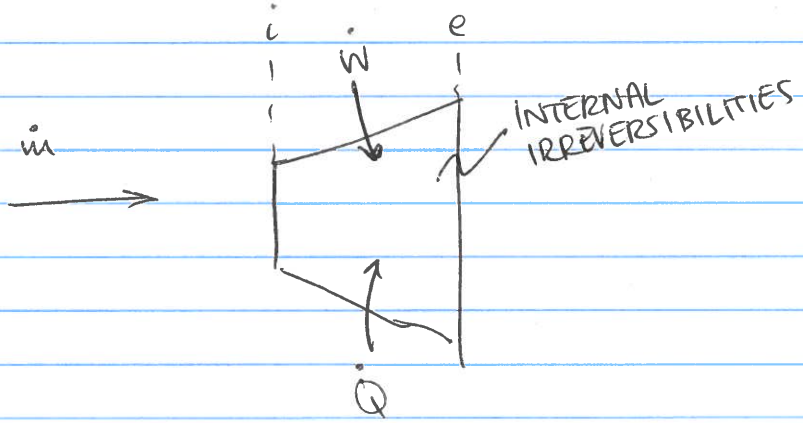
# Cycle Analysis of Jet Engines (Review)

Based on quasi-1D: flow direction thru flow path is aligned with x-axis. ($\vec{V} \rightarrow u$)

Consider engine / engine component (general)

$\dot{W}$ = shaft work rate across boundary

$\dot{Q}$ = heat rate across boundary




$P_t, T_t$ => total conditions

quick review (again) of Ptotal, Ttotal

* How do $P_t, T_t$ vary in duct? What drives change? Note static P, T, etc. change due to area change (even if $\dot{Q}=0$, $\dot{W}=0$, no irreversibilities).

Define work per mass (lower case) $w$

" heat " " " " $q$

$$w_{i\rightarrow e} = \frac{\dot{W}}{\dot{m}}, \quad q_{i\rightarrow e} = \frac{\dot{Q}}{\dot{m}}$$

$$\frac{P_{t,e}}{P_{t,i}} = \exp\left[\int_i^e \frac{\delta w}{RT_t} - \frac{S_{\text{irr / non ideal heat transfer}}}{R}\right]$$

where $\delta w$ = differential work interaction

$w_{i\rightarrow e} = \int_i^e \delta w$

NON-IDEAL HEAT INTERACTION OCCURS WHEN HEAT INTERACTION OCCURS AT A FINITE (NON-ZERO) MACH NUMBER ($T < T_t$)

$ds = \frac{\delta q}{T}$ $\Rightarrow$ MOST IDEAL TO ADD HEAT AT HIGHEST TEMP WHICH OCCURS AT LOWEST VELOCITIES, M

$$\frac{T_{t,e}}{T_{t,i}} = 1 + \frac{q_{i\to e} + W_{i\to e}}{C_p T_{t,i}}$$

FROM THE ENERGY EQUATION

$$C_p(T_{t,e} - T_{t,i}) = q_{i\to e} + W_{i\to e}$$

* THE DRIVERS:

- $P_t$ CHANGES OCCUR DUE TO WORK INTERACTIONS, INTERNAL IRREVERSIBILITIES, NON-IDEAL HEAT INTERACTION

- $T_t$ CHANGES OCCUR ONLY DUE TO ENERGY INTERACTIONS ACROSS BOUNDARY (NO DIFFERENCE BETWEEN HEAT AND WORK)

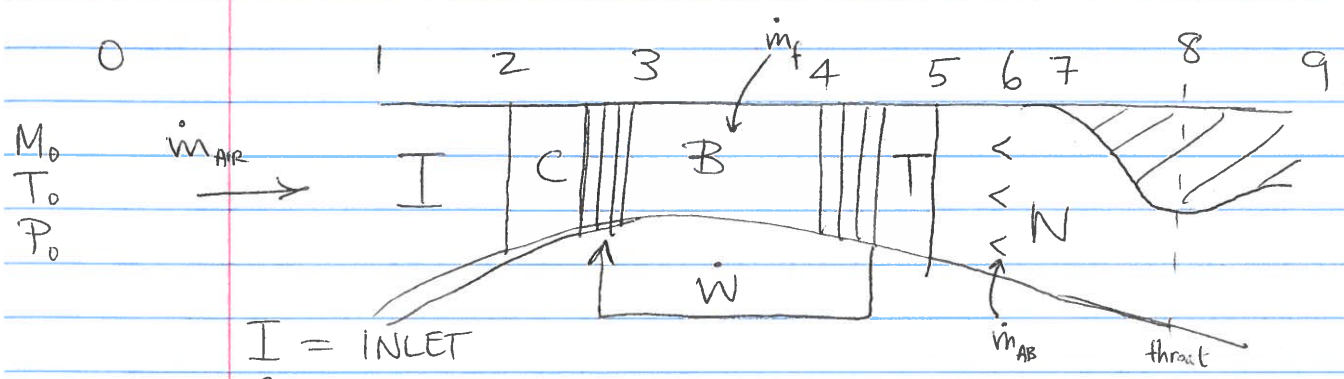
back to jet engine/ cycle analysis

DEFINE: $\pi = \frac{P_{t,e}}{P_{t,i}}$ ; $\tau = \frac{T_{t,e}}{T_{t,i}}$

* COMMONLY USED ENGINE SUBSCRIPTS:

- $d$ → DIFFUSER (INLET)
- $c$ → COMPRESSOR (MECHANICAL)
- $c'$ OR 'FAN' OR F → FAN (LOW PRESSURE COMPRESSOR)
- $b$ → BURNER
- $t$ → TURBINE
- $n$ → NOZZLE

# SCHEMATIC 'IDEAL' TURBOJET & NUMBERING (AIAA STANDARD)




I = INLET
C = COMPRESSOR
B = BURNER
T = TURBINE
N = NOZZLE

$$T_{t,0} = T_0 \left(1 + \frac{\gamma - 1}{2} M_0^2\right)$$

$T_{tA}$, $\Pi_C = \frac{P_{t3}}{P_{t2}}$ ARE GENERALLY PRESCRIBED

$h$ = 'HEATING VALUE' OF FUEL [J/kg]

## ☆ IDEAL TURBOJET (FLOWPATH RULES)

$\Pi_d = 1.0$, $\tau_d = 1.0$

$\Pi_c = \tau_c^{\frac{\gamma}{\gamma - 1}}$ ISENTROPIC

$\Pi_b = 1.0$, $\Pi_t = \tau_t^{\frac{\gamma}{\gamma - 1}}$, $\Pi_n = \tau_n = 1.0$

$\dot{m}_f = \dot{m}_{fuel}$,

WE HAVE TO HAVE $P_0 = P_9$ MATCHED

NO CONIDERATION OF ADDITIVE OR EXTERNAL DRAG.

ALSO, $\dot{m}_f \ll \dot{m}_{AIR}$

$f$ = FUEL-TO-AIR RATIO $= \frac{\dot{m}_f}{\dot{m}_{AIR}} \ll 1$

$f \approx 0.02$, $f_{STOICH} \cong 0.06$

ALSO, $c_p, \gamma, R \Rightarrow$ CONSTANT

## ANALYSIS OF AN IDEAL ENGINE

TURBINE – COMPRESSOR POWER BALANCE

$$\dot{m}_{AIR} c_p (T_{t3} - T_{t2}) = (\dot{m}_{AIR} + \dot{m}_f) c_p (T_{t4} - T_{t5})$$

BUT $\dot{m}_f \ll \dot{m}_{AIR}$ SO ...

$$\tau_t = \frac{T_{t5}}{T_{t4}} = 1 - \frac{\tau_r}{\tau_\lambda}(\tau_c - 1)$$

where $\tau_r = \frac{T_{t0}}{T_0} = \frac{T_{t2}}{T_0} = 1 + \frac{\gamma-1}{2} M_0^2$

AND $\tau_\lambda = \frac{c_{p4} T_{t4}}{c_{p0} T_0} = \frac{T_{t4}}{T_0}$

ALSO $\pi_r = \frac{P_{t0}}{P_0} = \left(1 + \frac{\gamma-1}{2} M_0^2\right)^{\frac{\gamma}{\gamma-1}}$

BURNER ENERGY EQUATION:

$$\dot{m}_f h = (\dot{m}_{AIR} + \dot{m}_f) c_p T_{t4} - \dot{m}_{AIR} c_p T_{t3}$$

$$\Rightarrow f = \frac{\dot{m}_f}{\dot{m}_{AIR}} = \frac{c_p T_0}{h} (\tau_\lambda - \tau_r \tau_c) \quad [\text{FOR } f \ll 1]$$

AE5535

## 2 ENGINE PERFORMANCE PARAMETERS :

a) SPECIFIC THRUST $= \dfrac{F}{\dot{m}_{AIR}}$

WHERE '$F$' IS UNINSTALLED THRUST

$$F = (\dot{m}_{AIR} + \dot{m}_f) u_9 - \dot{m}_9 u_0 + (P_9 - P_0) A_9$$

IDEALLY, $P_9 = P_0$ AND $f \ll 1$

$$\therefore F_{IDEAL} = \dot{m}_{AIR} (u_9 - u_0)$$

$$\therefore \frac{F_{IDEAL}}{\dot{m}_{AIR}} = u_9 - u_0$$

b) SPECIFIC FUEL CONSUMPTION $= S = \dfrac{\dot{m}_f}{F}$

$$\longrightarrow S = \frac{f}{(F/\dot{m})} \times 10^6 \quad \left[\frac{mg}{N \cdot sec}\right]$$

$$\left( \tau_\lambda = \frac{T_{t4}}{T_0} \, , \; \tau_r = \frac{T_{t0}}{T_0} = 1 + \frac{\gamma - 1}{2} M_0^2 \, , \cdots \right)$$

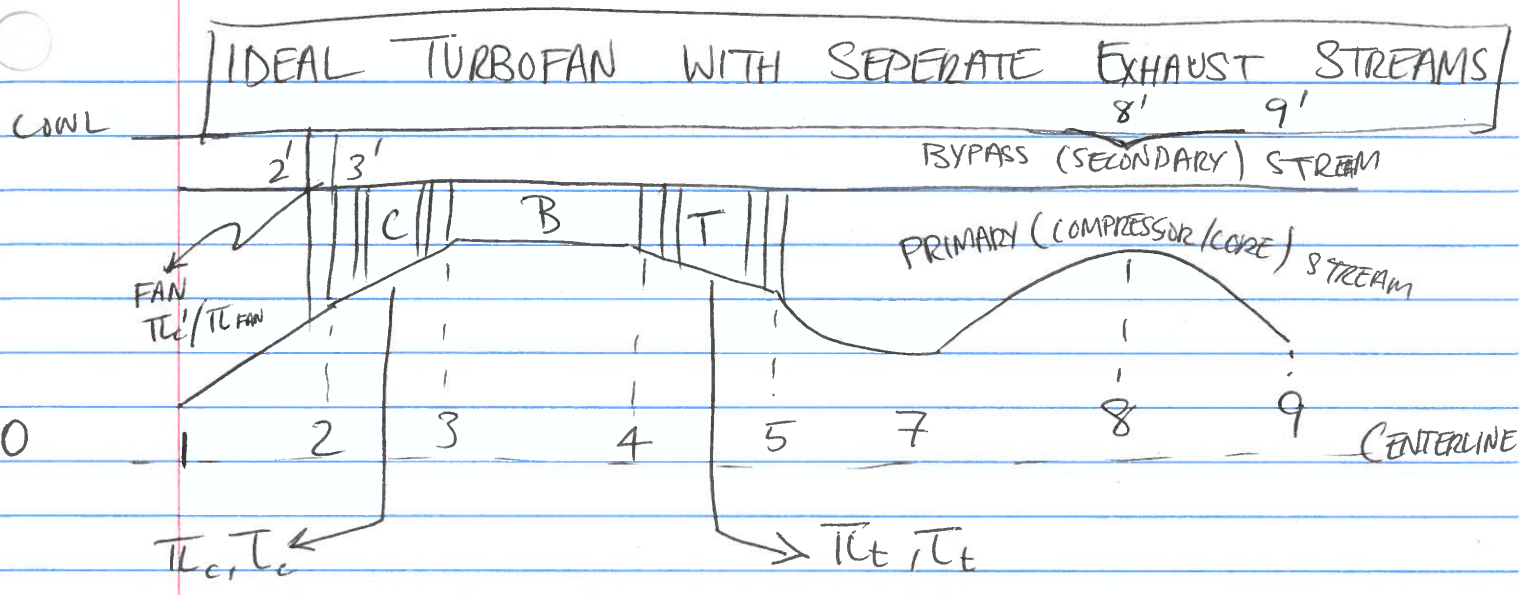
FOR INSTANCE, FOR IDEAL RAMJETS : $\pi_{t_c} = \tau_c = 1.0$

$\pi_{t_t} = \tau_t = 1.0$

$$\text{RAMJET} \dashrightarrow \frac{F}{\dot{m}} = a_0 M_0 \left[ \sqrt{\frac{\tau_\lambda}{\tau_r}} - 1 \right] \; : \; a_0 = \sqrt{\gamma R T_0}$$

* ALLOWS PARAMETRIC STUDIES OF ENGINES (i.e. FIX $M$, $T_{t4}$, $T_0$, $h$ AND VARY $\pi_{t_c}$ TO STUDY IMPACT OF COMPRESSOR SIZE ON $\frac{F}{\dot{m}}$, $S$)

# IDEAL TURBOFAN WITH SEPERATE EXHAUST STREAMS




- OUTER OR BYPASS STREAM DESIGNATED BY 'FAN', 'F', OR 'c'
- COMPRESSOR OR CORE STREAM DESIGNATED BY C (SOMETIMES 'P' FOR PRIMARY)

FOR THE IDEAL TURBOFAN:

$$T_{t_c}' = T_c'^{\frac{\gamma}{\gamma-1}} \quad \text{OR} \quad T_{t_{FAN}} = T_{FAN}^{\frac{\gamma}{\gamma-1}}$$

$$\alpha \text{ (OR } \beta) = \text{BYPASS RATIO} = \frac{\dot{m}_{FAN}}{\dot{m}_{COMPRESSOR}}$$

TOTAL CONDITIONS AT 2' SAME AS 2

TURBINE - COMPRESSOR - FAN POWER BALANCE:

$$\underbrace{(\dot{m}_c + \dot{m}_f) C_{p_4} T_{t_4} - (\dot{m}_c + \dot{m}_f) C_{p_5} T_{t_5}}_{\text{TURBINE POWER MAGNITUDE}}$$

$$= \underbrace{\dot{m}_c C_{p_0} (T_{t_3} - T_{t_2})}_{\text{COMPRESSOR POWER}} + \dot{m}_{FAN} C_{p_0} (T_{t_3'} - T_{t_2'})$$

$C_p$ = CONST FOR IDEAL

$$F_{\substack{\text{TURBO}\\\text{FAN (IDEAL)}}} = \dot{m}_c(u_9 - u_0) + (P_9 - P_0)A_9 + \dot{m}_F(u_9' - u_0) + (P_9' - P_0)A_9'$$

IDEAL Turbofan (see next page)

★ SOLUTION TO HANDOUT EXAMPLE: (DIRECT ANALYSIS)

a) $\dot{m}_f h = \dot{m}_c C_p (T_{t4} - T_{t3})$

b) $\dot{m}_c C_p (T_{t3} - T_{t2}) + \dot{m}_F C_p (T_{t3'} - T_{t2'}) = \dot{m}_c C_p (T_{t4} - T_{t5})$

Where $T_{t2} = T_{t2'}$

$\therefore$ $T_{t5} = T_{t4} - T_{t3} - 3 \cdot T_{t3'} + 4\, T_{t2}$

c) $P_{t2'} = P_{t0} = P_0 \left(1 + \frac{\gamma-1}{2} M_0^2\right)^{\frac{\gamma}{\gamma-1}} = 191{,}801\ P_a$

$T_{t0} = T_{t2'} = T_0 \left(1 + \frac{\gamma-1}{2} M_0^2\right) = 345.6\ K$

$P_{t3'} = P_{t2'}\, \Pi_{FAN} = 575\,403\ P_a = P_{t9'}$

$T_{t3'} = T_{t2'} (\Pi_{FAN})^{\frac{\gamma-1}{\gamma}} = 473\ K = T_{t9'}$

d) $P_{t2} = P_{t0} = P_0 \left(1 + \frac{\gamma-1}{2} M_0^2\right)^{\frac{\gamma}{\gamma-1}} = 191{,}801\ P_a$

$T_{t2} = T_{t0} = T_0 \left(1 + \frac{\gamma-1}{2} M_0^2\right) = 345.6\ K$

$P_{t3} = P_{t2}\, \Pi_C = 5\,754\,030\ P_a = P_{t4}$

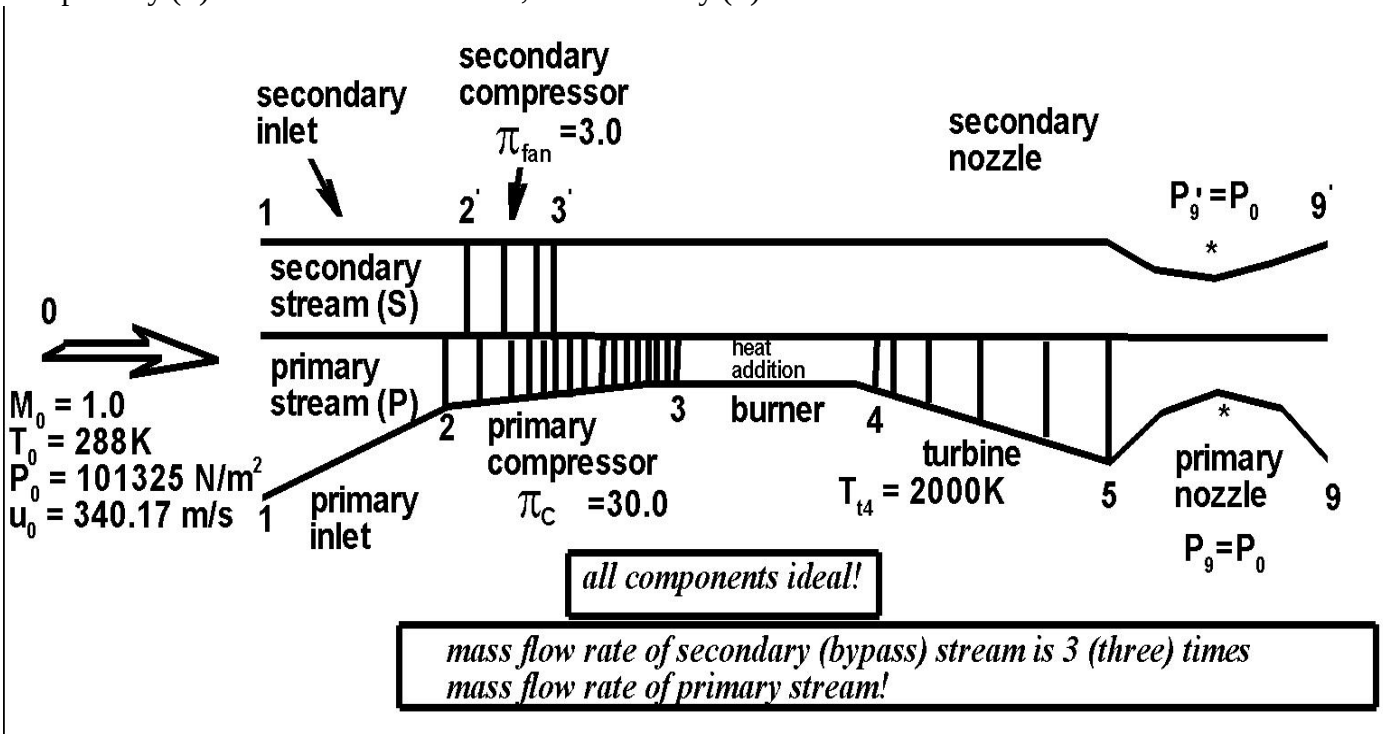
$T_{t3} = T_{t2} (\Pi_C)^{\frac{\gamma-1}{\gamma}} = 913.3\ K$

$T_{t5} = 1050\ K = T_{t9}$

$P_{t5} = P_{t4} \left(\frac{T_{t5}}{T_{t4}}\right)^{\frac{\gamma}{\gamma-1}} = 603\,275\ P_a = P_{t9}$

**AE 5535** **Review: Direct analysis of ideal engine (turbofan)**

Consider an engine composed of two streams; a primary (inner or core) stream and a secondary (outer or bypass) stream (see the sketch below). In the primary stream (often identified using a 'P' or sometimes a 'c' – for compressor) there is a compressor (stations 2 to 3), burner (3 to 4), and a turbine (4 to 5). In the secondary stream (designated 'fan' or sometimes 'S' or even c[1]) there is a fan which is a low-pressure ratio compressor (located from stations 2' to 3' (where the ' notation - the prime notation - on stations simply designate the secondary (S) stream). The primary (P) stream exits at station 9; the secondary (S) stream exits at station 9'.




The turbine supplies power to BOTH compressor and fan. All components are ideal (including ideally expanded nozzles). NOTE THAT THE AIR MASS FLOW RATE PROCESSED BY THE SECONDARY 'S' STREAM IS 3 TIMES (3X) THE AIR MASS FLOW RATE PROCESSED BY THE PRIMARY 'P' STREAM.

The engine is operating at flight Mach, $M_0 = 1.0$, $T_0 = 288K$, $P_0 = 101{,}325\ N/m^2$.

$$\frac{P_{t3'}}{P_{t2'}} = \pi_{fan} = 3.0 \qquad \frac{P_{t3}}{P_{t2}} = \pi_C = 30.0 \qquad T_{t4} = 2000K$$

$$h = 4.42x10^7\ J/kg(fuel) \qquad C_P = 1004.5\ J/kg-K \qquad R = 287\ J/kg-K$$

$$\gamma = 1.4 \qquad P_9 = P_{9'} = P_0 \qquad \dot{m}_{fan} = 3\dot{m}_{C(Primary)}$$

a) write the expression for the burner enthalpy balance (using *h* and mass flow rates of air and fuel)
b) write the expression for the power balance between the compressors and the turbine
c) calculate $P_{t2'}, T_{t2'}, P_{t3'}, T_{t3'}, P_{t9'}, T_{t9'}$ (i.e. total pressures, total temperatures in secondary stream, S)
d) calculate $P_{t2}, T_{t2}, P_{t3}, T_{t3}, P_{t4}, P_{t5}, T_{t5}, P_{t9}, T_{t9}$ (i.e. total pressure, total temperatures in primary stream, P)
e) find $M_9$ and $M_{9'}$
f) find $T_9$ and $T_{9'}$
g) find $u_9$ and $u_{9'}$
h) find the specific thrust of this engine
i) find the specific fuel consumption of this engine

* IT'S TYPICAL FOR A JET ENGINE TO HAVE TOTAL CONDITIONS AT EXHAUST THAT ARE HIGHER THAN FREE STREAM TOTAL CONDITIONS COMING IN.

e) $$\frac{P_{tq}}{P_q} = \frac{P_{tq}}{P_0} = \left(1 + \frac{\gamma-1}{2} M_q^2\right)^{\frac{\gamma}{\gamma-1}} \Rightarrow M_q = 1.823$$

$$\frac{P_{tq'}}{P_{q'}} = \frac{P_{tq'}}{P_0} = \left(1 + \frac{\gamma-1}{2} M_{q'}^2\right)^{\frac{\gamma}{\gamma-1}} \Rightarrow M_{q'} = 1.79$$

f) $$T_q = \frac{T_{tq}}{\left(1+\frac{\gamma-1}{2}M_q^2\right)} = 631\,K$$

$$T_{q'} = \frac{T_{tq'}}{\left(1+\frac{\gamma-1}{2}M_{q'}^2\right)} = 288\,K = T_0$$ AS EXPECTED FOR THIS CASE (IDEAL)

g) $$u_q = M_q\sqrt{\gamma R T_q} = 918\ m/s$$

$$u_{q'} = M_{q'}\sqrt{\gamma R T_{q'}} = 610\ m/s$$

h) $$\frac{F}{\dot{m}_{AIR,TOT}} = \frac{\dot{m}_c(u_q - u_0) + \dot{m}_F(u_{q'} - u_0)}{\dot{m}_c + \dot{m}_F}$$

$$\dot{m}_{AIR,TOT} = (\alpha+1)\dot{m}_c$$

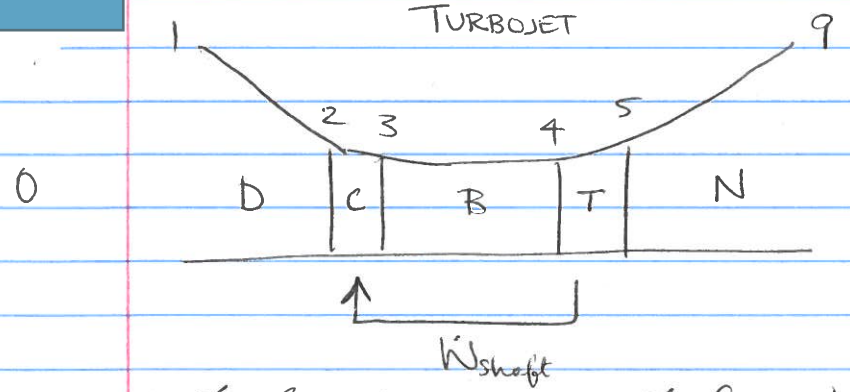
$$F/\dot{m}_{tot} = 346.6\ \frac{N\cdot s}{kg}$$

i) $$S = \frac{\dot{m}_f}{F} = \frac{\dot{m}_f/\dot{m}_{TOT}}{F/\dot{m}_{TOT}} = \frac{\dot{m}_f/\dot{m}_c}{(1+\alpha)\,F/\dot{m}_{tot}} = \frac{f}{(1+\alpha)\,F/\dot{m}_{tot}} \times 10^6$$

$$f = \frac{c_p(T_{t4} - T_{t3})}{h}$$

$$\therefore S = 17.82\ \frac{mg}{N\cdot s}$$

# NON-IDEAL ENGINE EFFECTS




R → DOESN'T CHANGE APPRECIABLY (FUEL LEA

COULD HAVE $\gamma_{AB}, C_{P_{AB}}$ IF WE HAVE AB

$P_9$ = NOZZLE EXIT PRESSURE $\neq P_0$

$\Rightarrow$ INCLUDE $P$ TERMS IN THRUST

DON'T ASSUME $\dot{m}_f \ll \dot{m}_a$ OR $f \ll 1$ (KEEP IT, THAT IS)

* INLET $\pi_{C_d} < 1.0$ , $\tau_d = 1.0$ [UNLESS VERY HIGH SPEEDS]
* COMPRESSOR/FAN $\pi_c = \tau_c^{\frac{e_c \gamma_c}{\gamma_c - 1}}$ : $e_c$ = POLYTROPIC COMPRESSOR EFFICIENCY

  OR $\tau_c = 1 + \dfrac{\pi_c^{\frac{\gamma_c - 1}{\gamma_c}} - 1}{\eta_c}$ : $\eta_c$ = GENERAL COMPRESSOR EFFICIENC

  $e_c'$, $\eta_c'$ FOR FAN

* BURNER $\pi_b < 1.0$ DUE TO IRREVERSIBILITIES AND NON-IDEAL HEA

  $\eta_b$ = BURNER EFFICIENCY $\quad \eta_b = \dfrac{(\dot{m}_a + \dot{m}_f) C_{P_t} T_{t_4} - \dot{m}_a C_{P_c} T_{t_3}}{\dot{m}_f h}$

* TURBINE $\pi_t = \tau_t^{\frac{\gamma_t}{(\gamma_t - 1) e_t}}$ : $e_t$ = POLYTROPIC TURBINE EFFICIENCY

  $\tau_t = 1 - \eta_t \left[ 1 - \pi_t^{\frac{\gamma_t - 1}{\gamma_t}} \right]$ : $\eta_t$ = GENERAL TURBINE EFFICIENC

* NOZZLE $\pi_{C_N} < 1.0$ , $\pi_N = 1.0$

MECHANICAL EFFICIENCY $\eta_M$ :
(DUE TO LOSSES) IN SHAFT WORK AND/OR AUXILARY POWER SUPPLY TO SUBSYSTEMS

$$\eta_M C_{P_t} (\dot{m}_a + \dot{m}_f)(T_{t4} - T_{t5}) = C_{P_e} \dot{m}_a (T_{t3} - T_{t2})$$

CHAIN $P_t, T_t$ THROUGH THE ENGINE TO GET $\frac{F}{\dot{m}}$ & S
IF $P_9$, $P_9'$ GIVEN ; FIND $M_9$ (ALSO KNOW $P_{t9}$, $T_{t9}$)

SUPPOSE $A_9$, $A_9'$, $\dot{m}_9$ GIVEN INSTEAD $\Rightarrow$ FIND $M_9$

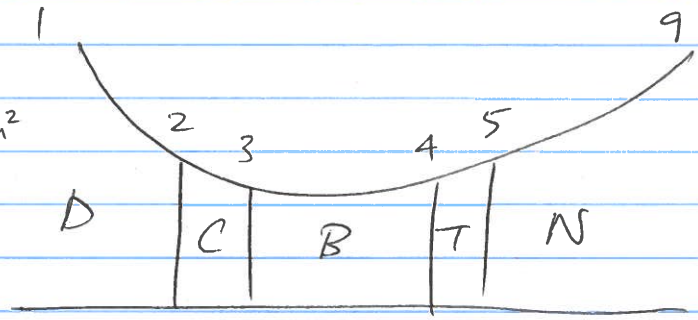
$$\dot{m} = \frac{A P_t}{\sqrt{T_t}} \sqrt{\frac{\gamma}{R}} M \left(1 + \frac{\gamma - 1}{2} M^2\right)^{\frac{-(\gamma+1)}{2(\gamma-1)}} \longrightarrow \text{FIND } M$$

IF $A_9 = A_8 = A_{throat}$ (ONLY CONVERGENT NOZZLE)
THEN $M_9 = M_8 = 1.0$

# EXAMPLE OF NON-IDEAL TURBOJET

$M_0 = 1.0$
$T_0 = 240K$
$P_0 = 10\,000\ N/m^2$

0 → RIGGING' ALTITUDE (DOESN'T CORRESPOND TO TABLES)




$\frac{P_9}{P_0} = 1.1$

**GIVEN**

$T_{t4} = 1800K$
$\pi_c = 20$
$\pi_d = .98$
$e_c = 0.9$
$\pi_b = .98$
$e_t = 0.9$
$\eta_M = 0.95$

$\pi_N = .99$
$h = 4.42 \times 10^7\ J/kg$
$C_{p_t} = 1098.2\ J/kg$
$C_{p_c} = 996.5\ J/kg$
$\gamma_t = 1.35$
$\gamma_c = 1.4$

**ANALYSIS:**

$$T_{t_0} = T_0\left(1 + \frac{\gamma-1}{2} M_0^2\right) = 288K$$

$$P_{t_0} = 18930\ Pa$$

$$\frac{T_{t3}}{T_{t2}} = \tau_c = \pi_c^{\frac{\gamma_c - 1}{\gamma_c e_c}} = 2.588$$

$$T_{t2} = T_{t0} = 288K \Rightarrow T_{t3} = T_{t2}\tau_c = 745.5K$$

$$P_{t_2} = P_{t_0}\pi_d = 18551\ Pa$$

$$P_{t_3} = P_{t_2}\pi_c = 371014\ Pa$$

BURNER ENERGY BALANCE: (ENTHALPY)

$$(\dot{m}_a + \dot{m}_f) C_{p_t} T_{t4} - \dot{m}_a C_{p_c} T_{t3} = \dot{m}_f h \eta_b \quad \text{DIVIDE BY } \dot{m}_a$$

$$f = \frac{\dot{m}_f}{\dot{m}_a} = \frac{C_{p_t} T_{t4} - C_{p_c} T_{t3}}{\eta_b h - C_{p_t} T_{t4}} = 0.0298$$

$$P_{t4} = P_{t3}\pi_b = 363594\ Pa$$

AE 5535

TURBINE/COMPRESSOR POWER BALANCE:

$$\dot{m}_a C_{p_c}(T_{t3}-T_{t2}) = (\dot{m}_a+\dot{m}_f) C_{p_t}(T_{t4}-T_{t5})\eta_M$$

$$\Rightarrow T_{t5} = T_{t4} - \frac{C_{p_c}(T_{t3}-T_{t2})}{\eta_M (1+f) C_{p_t}} = 1375.7\ K$$

$$\left(\frac{P_{t5}}{P_{t4}}\right) = \left(\frac{T_{t5}}{T_{t4}}\right)^{\frac{\gamma_t}{(\gamma_t-1)e_t}} \Rightarrow \pi_{t_t} = 0.316$$

$$P_{t5} = \pi_b P_{t4} = 114\,893\ Pa$$

$$P_{tq} = P_{t5}\pi_N = 113\,744\ Pa$$

$$T_{tq} = T_{t5}\tau_N = T_{t5} = 1375.7K$$

$$P_q = P_0(1.1) = 11\,000\ Pa$$

$$\frac{P_{tq}}{P_q} = \left(1+\frac{\gamma_t-1}{2}M_q^2\right)^{\frac{\gamma_t}{\gamma_t-1}} \Rightarrow M_q = 2.181$$

$$T_q = T_{tq}\left(1+\frac{\gamma_t-1}{2}M_q^2\right)^{-1} = 750.76K$$

$$u_q = M_q\sqrt{\gamma_t R T_q} = 1171.6\ m/s$$

$$\frac{F_x}{\dot{m}_a} = (1+f)u_q - u_0 + \frac{(P_q-P_0)A_q}{\dot{m}_a}$$

where $\dfrac{(P_q-P_0)A_q}{\dot{m}_a} = \dfrac{(P_q-P_0)}{\rho_q}$ ---- ↓

$$\left(1-\frac{P_0}{P_q}\right)\frac{\rho_q R T_q A_q u_q}{\dot{m}_a u_q} \cdots \left(1-\frac{P_0}{P_q}\right)(1+f)R\frac{T_q}{u_q}$$

$$\therefore \frac{F}{\dot{m}_a} = 914.4\ \frac{N\cdot s}{kg}$$

$$S = \frac{f\times 10^6}{F/\dot{m}_a} = 32.6\ \frac{mg}{N\cdot s}$$

FOR TURBOFANS :

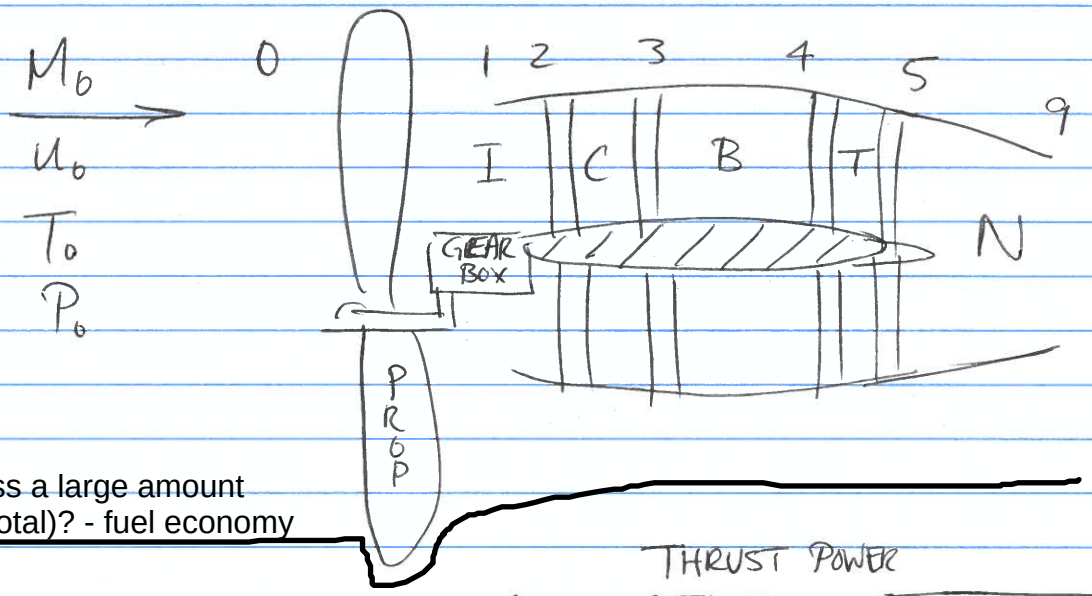
WE CAN FIND AN ABSOLUTE MINIMUM SPECIFIC FUEL CONSUMPTION.

IN $(\pi_{c_1}, \pi_{c_2}, \alpha)$ SPACE

FIND BET COMBINATION (OPTIMIZATION)

AE5535

# TURBOPROP ANALYSIS




reminder: why process a large amount of air (large mdot air total)? - fuel economy

* PROPULSIVE EFFICIENCY $= \eta_p =$ $\frac{\text{THRUST POWER}}{\text{RATE CHANGE IN KE ACROSS ENGINE}}$

$$\eta_p = \frac{F_x u_0}{\frac{1}{2}\dot{m} u_e^2 - \frac{1}{2}\dot{m} u_0^2}$$

(RECALL: $F_x = \dot{m}(u_9 - u_0)$ SUBSONIC)

WITH ASSUMPTIONS $\eta_p = \dfrac{2u_0}{u_9 + u_0}$ IMPLIES: $u_9 \to u_0$, $\eta_p \to 1.0$

WITH A FIXED $F_x$ REQUIREMENT $\longrightarrow$ WE WANT TO PROCESS LARGE $\dot{m}$ IF $u_9 \approx u_0$

* PROPELLER EFFICIENCY $= \eta_{prop} =$ $\frac{F_x u_0 \text{ (THRUST POWER DELIVERED by propeller)}}{\text{SHAFT POWER TO PROP (MECHANICAL FROM GEAR BOX)}}$

$$\eta_{prop} = \frac{F_x u_0}{\dot{W}_{prop}}$$

$F_x$: propeller thrust

NOTES:

* SOME TIMES (MOST OF THE TIME), TURBINE HAS 'SPOOLING' SUCH THAT A SEPERATE SEGMENT OF THE TURBINE RUNS THE PROP (COMMON FOR GAS TURBINES IN GENERAL FOR REDUNDANCY, STATIC STARTING)
* DUE TO THE LARGE DIAMETER OF PROPS, IT MUST GENERALLY ROTATE AT LOWR RPMs THAN CORE SHAFT. TO MINIMZE BLADE STRESSES ESPECIALLY WHEN IT REACHES TRANSONIC → WE NEED GEAR BOX.
* TURBOPROPS LIMITED TO $M_0 \lesssim 0.6$ OR 0.7 DUE TO BLADE TIP SPEEDS.
* CAN BE VERY FUEL EFFICIENT.
* TURBOSHAFT.

## CYCLE ANALYSIS NON-IDEAL TURBOPROP (SINGLE-SPOOL

THRUST → HAS TWO CONTRIBUTIONS, ONE FROM CORE STREAM, ONE FROM PROP STREAM.

a) CORE STREAM WILL HAVE SAME EXPRESSION FOR as turbojet

$\frac{F_{core}}{\dot{m}_{core}}$ / ~~[illegible]~~ AS OF $\frac{F_{core}}{\dot{m}_{tot}}$ ($f_{DF}$ ~~[illegible]~~ TURBOJET)

turboprop $\frac{F_{core}}{\dot{m}_c} = a_0\left[(1+f)M_0\frac{u_9}{u_0} - M_0\right]$ ASSUME $P_9 = P_0$ (NOT BAD ASSUMPTION)

$\frac{\dot{m}_f}{\dot{m}_c} = f = \frac{\tau_\lambda - \tau_r\tau_c}{\frac{\eta_b h}{c_{pc}T_0} - \tau_\lambda}$ $\quad \therefore \tau_\lambda = \frac{T_{t4}c_{pt}}{T_0 c_{pc}}$

$\tau_r = 1 + \frac{\gamma_c - 1}{2}M_0^2$

$$\pi_r = \tau_r^{\frac{\gamma_c}{\gamma_c - 1}} \quad , \quad \tau_c = \pi_c^{\frac{\gamma_c - 1}{\gamma_c e_c}}$$

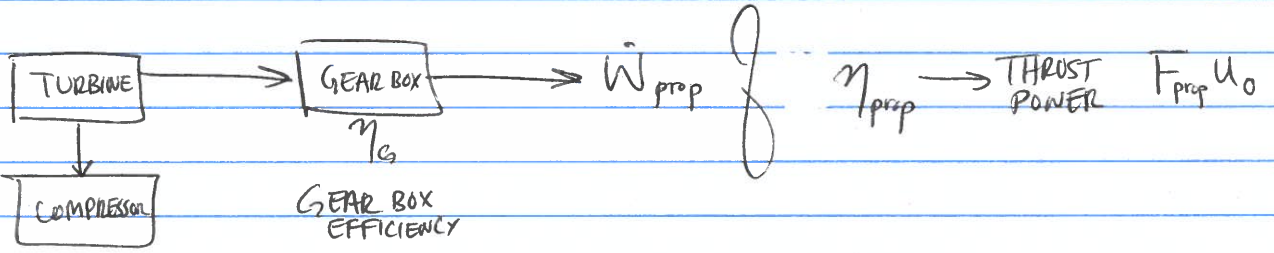
FROM THE TURBOJET ANALYSIS (CHAIN $P_t$, $T_t$ THROUGH) CAN SHOW:

$$\boxed{M_0 \frac{u_9}{u_0} = \left\{ \frac{2\tau_\lambda}{\gamma_c - 1} \left[ \tau_t - \frac{\tau_t^{\frac{-(1-e_t)}{e_t}}}{(\pi_r \pi_d \pi_c \pi_b \pi_n)^{\frac{\gamma_t - 1}{\gamma_t}}} \right] \right\}^{1/2}}$$

↑ DUE TO 'RAYLEIGH LOSS'

b) PROPELLER STREAM : FIND $\dot{W}_{prop}$

TURBINE SPLITS POWER BETWEEN GEARBOX AND COMPRESSO




POWER TO GEARBOX:

$$\dot{m}_c (1+f) \eta_m c_{p_t} (T_{t4} - T_{t5}) - \dot{m}_c c_{p_c} (T_{t3} - T_{t2})$$

TURBINE PWR MAGNITUDE — COMP PWR

$$\dot{W}_{prop} = \dot{m}_c c_{p_c} T_0 \eta_G \left[ \eta_m (1+f) \tau_\lambda (1 - \tau_t) - \tau_r (\tau_c - 1) \right]$$

Then $\eta_{prop} = \frac{F u_0}{\dot{W}_{prop}} \Rightarrow F u_0 = \eta_{prop} \dot{W}_{prop}$

$$\frac{F_{prop}}{\dot{m}_c} = \frac{\eta_{prop} \eta_G c_{p_c} T_0 \left[ \eta_m (1+f) \tau_\lambda (1 - \tau_t) - \tau_r (\tau_c - 1) \right]}{u_0}$$

at $u_0 = 0$, $\eta_{prop} = 0$ so $\frac{F_{prop}}{\dot{m}_c}$ DOESN'T GO TO $\infty$ AT $u_0$

$$\boxed{\left( \frac{F}{\dot{m}_{core}} \right)_{tot} = \frac{F_{core}}{\dot{m}_c} + \frac{F_{prop}}{\dot{m}_c}} \Rightarrow \boxed{S = \frac{f}{(F/\dot{m}_c)_{tot}}}$$

$\tau_t$ IS EITHER PROVIDED OR IS FOUND BY REQUIRING $S$ TO BE MINIMAL.

SET $\frac{\partial S}{\partial \tau_t} = 0$ TO GET $\tau_t^*$

WHICH IS $\tau_t$ AT MINIMUM $S$

DO THIS AND OBTAIN ($\gamma = 1.4$):

turbine total temp ratio that gives minimum S (specific fuel consumption)

$$\tau_t^* = \left\{ \frac{1}{K} \tau_t^{*\frac{e_t - 1}{e_t}} \right\} + \left\{ \frac{M_0^2 \left[ 1 + \frac{(1-e_t)}{e_t} \frac{1}{K} \tau_t^{*-\frac{1}{e_t}} \right]^2}{5 \tau_\lambda (\eta_{prop} \eta_G \eta_m)^2} \right\}$$

WHERE $K = (\tau_r \tau_d \tau_c \tau_b \tau_n)^{\frac{\gamma_t - 1}{\gamma_t}}$

YOU NEED TO FIND $\tau_t^*$ THAT SATISFIES THE EQN.

---

THE BYPASS RATIO $\alpha$ DOES NOT DIRECTLY APPEAR BECAUSE ONCE $\tau_t$ IS DETERMINED, THEN THE WORK SPLIT BETWEEN THE COMPRESSOR AND PROPELLER IS FIXED AND THE SIZE OF THE PROPELLER (DIAMETER, NUMBER OF BLADES, etc.) DETERMINE $\alpha$.

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☆ OPTIMIZED TURBOFANS CAN HAVE ~15% HIGHER $S$ THAN AN OPTIMIZED TURBOPROP FOR LOW SUBSONIC FLIGHT.

AE 5535

## EVALUATING $\frac{P_9}{P_0}$ OR $\frac{P_9'}{P_0}$ IN CYCLE ANALYSIS

IN USUAL SUBSONIC TRANSPORT APPLICATIONS WITH TURBOJET OR SEPERATE STREAM TURBOFAN OR TURBOPROP WITH NO AFTERBURNER, YOU CAN OFTEN GET AWAY WITH A SIMPLE CONVERGING NOZZLE SUCH THAT $M_9$ OR $M_9'$ ARE UNITY ($M_9 = 1.00$)

IF THAT'S TRUE, BY DEFINITION $\frac{P_{t9}}{P_9} = \left(\frac{\gamma_t+1}{2}\right)^{\frac{\gamma_t}{\gamma_t-1}}$

AND $\frac{P_{t9}'}{P_9'} = \left(\frac{\gamma_c+1}{2}\right)^{\frac{\gamma_c}{\gamma_c-1}}$

AND THEN YOU KNOW $\frac{P_{t9}}{P_0} = \frac{P_{t0}}{P_0}\frac{P_{t2}}{P_{t0}}\frac{P_{t3}}{P_{t2}}\frac{P_{t4}}{P_{t3}}\frac{P_{t5}}{P_{t4}}\frac{P_{t9}}{P_{t5}}$

$$\therefore \frac{P_{t9}}{P_0} = \pi_r \pi_d \pi_c \pi_b \pi_t \pi_n$$

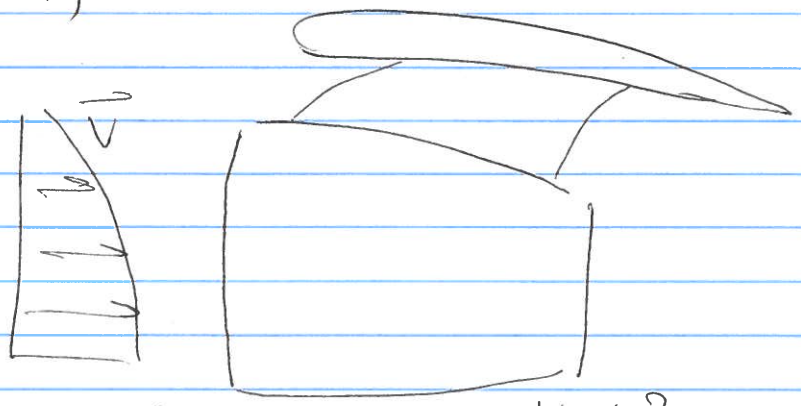
$$\frac{P_{t9}'}{P_0} = \pi_r \pi_d \pi_c' \pi_n'$$

$$\text{So } \frac{P_9}{P_0} = \frac{\pi_r \pi_d \pi_c \pi_b \pi_t \pi_n}{\left(\frac{\gamma_t+1}{2}\right)^{\frac{\gamma_t}{\gamma_t-1}}}$$

$$\frac{P_9'}{P_0} = \frac{\pi_r \pi_d \pi_c' \pi_n'}{\left(\frac{\gamma_c+1}{2}\right)^{\frac{\gamma_c}{\gamma_c-1}}}$$

# ONE-DIMESIONALIZATION OF 3-D FLOW FIELDS

(FLOW AVERAGING)




IF WE WANT TO TAKE A 3D FLOW (SAY FROM CFD) AND PUT IT THROUGH CYCLE ANALYSIS, WE WANT TO ONE-DIMENSIONALIZE IT. HOW?
HOW TO OBTAIN THE BEST 1D PARAMETERS?
i.e. DESIRE 1-D 'EQUIVALENT' FLOW FIELD.
— TO USE FLUX-BASED FLOW RATE CONSERVATION PRINCIPLES / METHODS.
SOLVE FOR $\rho_0, u_0, P_0, T_0$ WHERE '0' SUBSCRIPT DESIGNATES 1-D PARAMETER

SO AT STATION OF INTEREST, MASS FLOW RATE:

CONTINUITY

$$\int_A \rho u \, dA = \rho_0 u_0 A$$

KNOWN (3-D)

MOMENTUM

$$\int_A (\rho u^2 + P) \, dA = (\rho_0 u_0^2 + P_0) A$$

ENERGY (TOTAL ENTHALPY CONSERVED)

$$\int_A \rho u \left(C_p T + \frac{u^2}{2}\right) dA = \rho_0 u_0 \left(C_p T_0 + \frac{u_0^2}{2}\right) A$$

TO CLOSE THE SYSTEM

$$P_0 = \rho_0 R T_0$$

∴ 4 EQUATIONS, 4 UNKNOWNS, FIND: $\rho_0, u_0, P_0, T_0$

AND YOU ARE READY TO GO!

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ORDER OF PREFERENCE : OF ONE DIMENSIONALIZING FLOW

(1) PREVIOUS METHOD (OR BETTER YET, DISTORTION METHODOLOGY)
(2) MASS-AVERAGED $\cdots u_0 = \int \rho u^2 dA \,/\, \int \rho u dA$

OR AREA-AVERAGED
(3) OTHER

[redacted]